



A rationally designed polymer brush/lubricant coating system for effective static and dynamic marine antifouling

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ABSTRACT

Bioinspired slippery surfaces have garnered significant attention as promising solutions to mitigate biofouling. Unlike traditional lubricant-infused porous surfaces, recent research has focused on the integration of lubricants within polymer brush-grafted surfaces. The combination of polydimethylsiloxane (PDMS) polymer brush and silicone oil has gradually become the most prevalent choice due to their outstanding chemical affinity. However, this conventional coating system also has the potential for improvement to meet the needs of long-lasting and efficient marine antifouling, including i) precisely designing the polymer brush's chemical structure to match the polarity of a specific lubricant enhances their chemical affinity, and ii) appropriately reduce the coating system's surface energy to improve the fouling desorption performance. Here, we introduce a systematically engineered polymer brush/lubricant coating system that incorporates fluorinated polysiloxane and perfluoropolyether fluid. This novel coating system exhibits enhanced adhesion strength coupled with reduced surface energy, resulting in superior stability and omniphobic properties. Additionally, it showcases excellent corrosion resistance and significantly deters marine microorganism adhesion, achieving reductions of 98.8 % for *P. tricornutum* and 99.8 % for *Bacillus* sp.. Marine field trials, conducted over a 90-day static immersion period, confirm the remarkable antifouling performance, which surpasses most existing slippery coatings. Moreover, under dynamic conditions, organisms adhering for 150 days are readily dislodged by shear force. These findings underscore the pivotal role of systematic design in polymer brush/lubricant coating systems for the advancement of high-performance slippery surfaces tailored for marine antifouling applications.

1. Introduction

Marine biofouling, the undesirable attachment and proliferation of organisms on surfaces submerged in seawater, stands as a pressing global issue within the marine industry [1,2]. Its consequences include heightened hydrodynamic drag and fuel consumption for ships, accelerated material corrosion, reduced lifespan of marine structures, obstruction of seawater pipelines, and the potential for biological invasions [3–6]. Consequently, biofouling prevention holds substantial

practical importance, prompting extensive efforts towards the development of anti-biofouling surfaces over recent decades [7,8].

Natural antifouling surfaces provide valuable inspiration for the development of effective solutions against biofouling [5,6,9–11]. Inspired by the slippery inner walls of Nepenthes pitchers, the slippery liquid-infused porous surface (SLIPS) has emerged as a promising strategy for combating biofouling [12,13], garnering significant attention in marine anti-biofouling applications [14,15]. As illustrated in Fig. 1a, unlike solid surfaces typically considered favorable for

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settlement, the dynamic nature of the slippery liquid layer hinders fouling organisms from sensing and adhering [16]. Moreover, utilizing low surface tension lubricants can minimize adhesion strength by reducing strong polar interactions with biofoulants (e.g., hydrogen or ionic bonding), thereby facilitating easy detachment [17]. However, in SLIPS systems, the micro/nano-porous substrate merely acts as a reservoir for lubricants; once these lubricants are depleted, and the solid surface is exposed, the antifouling efficacy is completely compromised [18].

Recent studies have focused on integrating silicone oil into PDMS brush-grafted surfaces to develop slippery liquid coatings on flat surfaces [19–24]. In such coating systems, the grafted PDMS brush fulfills dual roles: i) it hosts and stabilizes silicone oil owing to their mutual affinity, and ii) it forms a liquid-like, omniphobic layer because of the highly flexible and mobile polymer chains, thereby exhibiting inherent slippery and antifouling properties [25,26]. Furthermore, the covalently grafted liquid-like layer exhibits greater robustness compared to physically adsorbed lubricants. Consequently, while the latter gradually depletes over time, the former is anticipated to persistently resist the adhesion of fouling organisms [27,28]. Thus, the judicious combination of polymer brush and lubricant can effectively enhance the antifouling durability of slippery liquid coatings.

However, upon employing the Owens-Wendt-Rabel-Kaelble (OWRK) method [29] to scrutinize the PDMS/silicone oil coating system, an unexpected revelation emerged: it fell outside the optimal adhesion corridor, as depicted in Fig. S1. Notably, despite theoretical non-polarity, a discernible polar component ranging from 2 % to 4 % is detectable on PDMS (Tables S1 and 2) [30–33,41]. This partial polarity

may compromise the chemical affinity of PDMS with non-polar silicone oil [19,34,35]. Therefore, the coating system has the potential for improvement. To systematically design a polymer brush/lubricant coating system for long-lasting and efficient marine antifouling, the following criteria should be considered: i) selection of a lubricant with low surface tension (γ_l) to facilitate efficient fouling desorption by the outermost liquid layer [36]; ii) incorporation of a liquid-like polymer layer with low surface energy (γ_s) to ensure prolonged antifouling efficacy by the grafted polymer brush [37]; and iii) establishment of strong adhesion between the lubricant and polymer brush to enhance overall system stability, as determined by the work of adhesion (W_A).

Based on these considerations, we have thoroughly examined the properties of common polysiloxanes and lubricants, leading to the discovery of an improved system compared to PDMS/silicone oil: the combination of poly[methyl (3,3,3-trifluoropropyl) siloxane] (PMTFPS, $\gamma_s^p = 0.52$ mN/m) with the perfluoropolyether fluid Krytox GPL 103 (K103, $\gamma_l^p = 0.50$ mN/m), as depicted in Fig. 1b. In this study, we fabricate the PMTFPS/K103 coating system on smooth titanium alloy surfaces and evaluate its liquid repellency, stability and antifouling performance. The coating system exhibits superior omniphobic properties and outstanding corrosion resistance. Additionally, it shows excellent stability in both static immersion and dynamic scouring. Laboratory experiments and marine field tests corroborate its high antifouling efficiency and long-term antifouling performance. Consequently, our findings position the PMTFPS/K103 coating system as a promising marine antifouling solution, offering inspiration for the systematic development of high-performance slippery liquid surfaces for

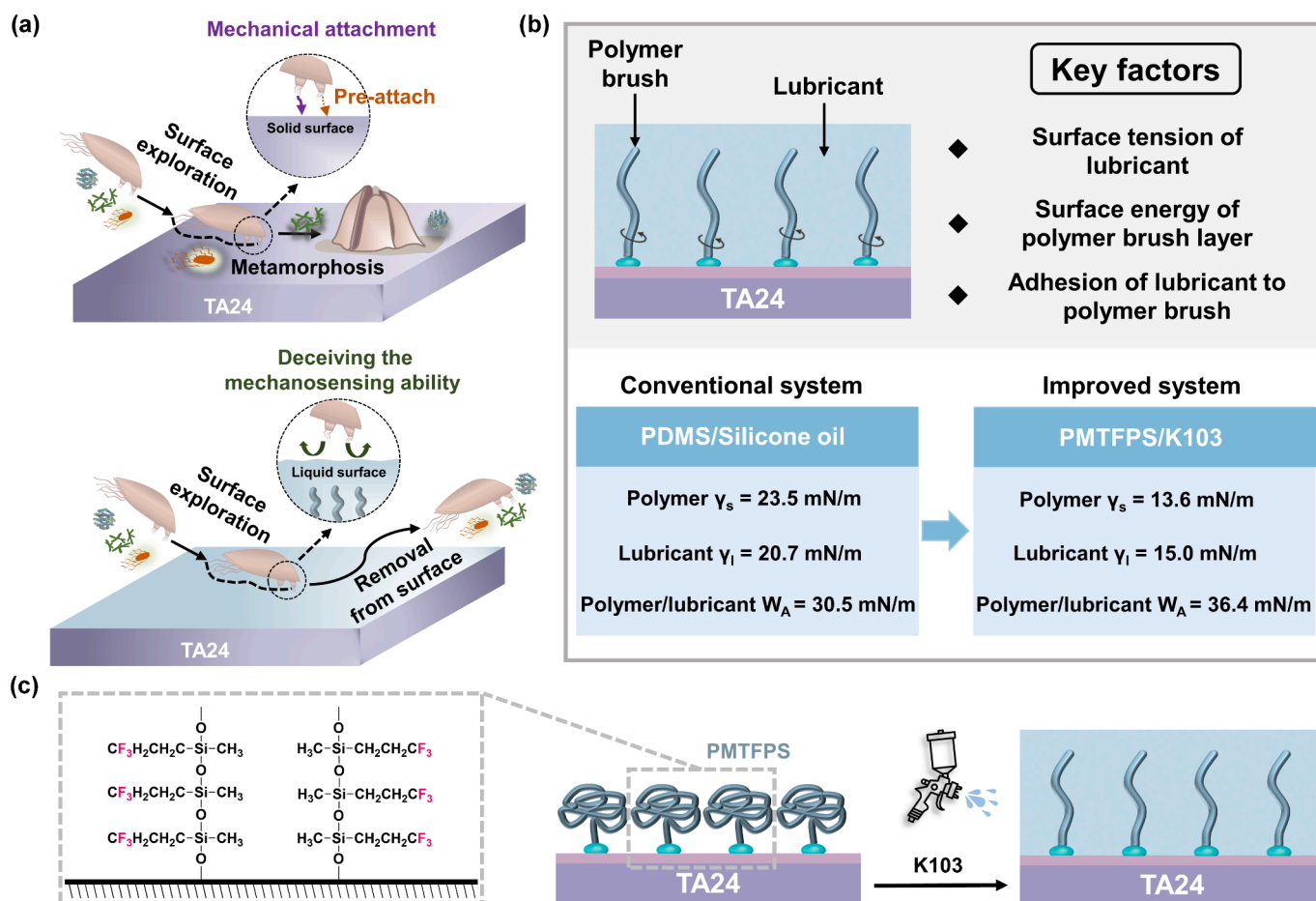


Fig. 1. A rationally designed PMTFPS/K103 coating system for antifouling. (a) The process of fouling organisms attaching to solid surfaces and the antifouling mechanism of liquid surfaces. (b) A rational strategy for polymer brush/lubricant coating system design. (c) Schematic illustration of the PMTFPS/K103 coating system preparation process.

practical antifouling applications.

2. Materials and methods

2.1. Materials

The titanium alloys (TA24) were purchased from Luoyang Sunrui Titanium Precision Casting Co., Ltd. The specimens were cut to specified sizes and subsequently ground with SiC paper, cleaned with deionized water and ethanol, and dried under a nitrogen stream. K103 was supplied by Chemours. Dimethoxy(methyl)(3,3,3-trifluoropropyl)silane (MTFPS) was purchased from Sigma Aldrich. Isopropyl alcohol, diiodomethane, ethylene glycol, and sulfuric acid were obtained from Aladdin Industrial Co., Ltd. Deionized water was supplied via an ultrapure water system (Direct-Q 3UV). All the chemical reagents were used as they were.

2.2. Preparation of PMTFPS polymer brush layer

This preparation was carried out at scales ranging from 20 to 100 g in clean glass screw-cap bottles. For example, 89 g of isopropanol, 10 g of MTFPS, and 1 g of sulfuric acid were added to a bottle. The solution was swirled to mix for about 30 s at room temperature before use. The substrate surface was treated by low temperature plasma treatment instrument (CAS PLASMA, CPT-CCP-0004LQX-A, China). The inlet gas was taken as water vapor carried by argon and treated for 4 min at 200 W power. Then the substrates were put into the reaction solution for 30 s and removed. Excess liquid was drained by brief contact with filter paper. The substrate was allowed to dry for 30 min at room temperature. Deionized water and isopropanol were used to rinse the surface and blow dry. The surface of the substrate gradually became hydrophobic during the rinsing process, and the PMTFPS polymer brush layer was successfully prepared.

2.3. Preparation of PMTFPS/K103 coating system

The K103 was used as the lubricant, applied evenly to the surface of the PMTFPS polymer brush layer by spray gun, and after inclined placement natural leveling. The PMTFPS/K103 coating system was prepared on the surface of TA24 with the PMTFPS polymer brush layer.

2.4. Simulation of the bonding strength between polymer brush monomer and substrate

The molecular dynamic simulation was conducted on MATERIALS AND PROCESSES SIMULATIONS (MAPS) 4.3.0. The Amorphous Cell, Polymer builder, Crystal builder, Surface builder, and LAMMPS modules were utilized and the force field of PCFF was employed. The interaction energy between the monomer of the polymer brush and the TA24 substrate surface was analyzed.

2.5. Calculation of grafting density

The dimensionless grafting density $\Sigma = \sigma \pi R_g^2$, where grafting density σ only represents the areal density of chains and does not account for their packing behavior (whether they overlap or not) on the surface ($\sigma = (h\rho N_A)/M_w$, h is polymer layer thickness, ρ is polymer monomer density, N_A is Avogadro's number and M_w is the molecular weight of the polymer chains), R_g is the polymer chain's gyration radius. MAPS was used to construct the optimal model of polymer brush to simulate and calculate parameters such as R_g when forming different thicknesses of polymer brush layers.

2.6. The work of adhesion analysis

The W_A between the coating system and the substrate with polymer

brush layer was measured by the OWRK method with the Theta Flex CA meter. Water, ethylene glycol, and diiodomethane were used as reference liquids. The sessile drop method was employed to evaluate the CAs of reference liquids. For the W_A calculation, the surface tension with disperse γ^d and polar γ^p contributions of the reference liquids was obtained from the Theta Flex software.

2.7. Anti-diatom assays

P. tricornutum was selected as the target diatom. Settlement assays of diatom cells were conducted on TA24 substrate without any modification (as a control), grafted with PMTFPS and K103 as the reference. The f/2 nutrient salts ($\text{Na}_2\text{SiO}_3 \cdot 9\text{H}_2\text{O}$, NaNO_3 , NaH_2PO_4 , Vitamins, Trace metal) were added 1 mL each into 1000 mL sterilized artificial seawater for the culture of *P. tricornutum*. The different substrates were immersed in a petri dish containing 30 mL of *P. tricornutum* (5×10^6 cells/mL), respectively, and grew in a light incubator at 20 °C with 12 h light and 12 h dark cycle for 7 days. After 7 days, the unattached diatoms on the surfaces were removed three times with sterilized artificial seawater, and the substrates were fixed with 2.5 % glutaraldehyde at 4 °C for 4 h. After the surfaces were fixed with glutaraldehyde, the substrates were rinsed with deionized water for three times. The adhesion of *P. tricornutum* was observed by a laser confocal scanning microscope (LCSM, Leica TCS SP8), and the adhesion rate was calculated by Image J software.

2.8. Antibacterial assays

In this study, *Bacillus* sp. was selected as model bacteria. *Bacillus* sp. was inoculated into 2216E liquid medium and cultured to an OD value of 1.0 at 600 nm in a constant temperature shaking table of 150 rpm at 30 °C, and the bacterial density was 10^9 cfu/mL. Then the 2216E liquid medium was diluted to 1×10^6 cfu/mL. After ultraviolet sterilization, the prepared substrates were placed in a petri dish, and the diluted bacterial solution was added, and incubated in a constant temperature incubator at 30 °C for 72 h. After culture, the non-adherent *Bacillus* sp. was removed by washing with sterilized artificial seawater three times, and then fixed with 2.5 % glutaraldehyde at 4 °C for 4 h. After fixation with glutaraldehyde, the surfaces were cleaned with deionized water three times, and then stained with 50 mg/L propyl iodide under dark conditions for 15 min, and then cleaned with deionized water to remove excess propyl iodide. The stained surfaces were observed under a confocal laser scanning microscope (CLSM, Leica TCS SP8) to capture images of the surfaces at different locations. At least three parallel experiments were conducted for each surface, and the adhesion rate was calculated using Image J software.

2.9. Ocean hanging board test

In the natural seawater conditions, the hanging board test was carried out in the floating raft in the sea area of Xiamen, where a wide variety of fouling organisms were tested. In the test of the hanging board, TA24 was used as the substrate, with the size of 100 mm $\times$ 100 mm $\times$ 5 mm. The test panels were fixed to the hanging bracket using copper wire, and the frame was installed 1 m below the surface of the sea. The panels were taken regularly, and photos were taken to record the biological contamination of the surface. During this time, the adhesion of marine fouling organisms on the surface can be observed to evaluate the actual anti-fouling effect of the coating system.

2.10. Characterization

The morphology of TA24 was characterized by high-resolution field-emission scanning electron microscopy (Zeiss, Sigma 500, Germany). The element distributions on the TA24 substrate were obtained by an energy-dispersive spectrometer (Zeiss, Sigma 500, Germany). X-ray

photoelectron spectroscopy (XPS) was carried out by using an electron spectrometer (Thermo escalab 250Xi, USA) with an Al Ka X-ray source ($h\nu = 1486.6$ eV, the power was 150 W). The wettability of different liquids on TA24 surfaces was measured by sessile drop methods on the CA meter (Biolin, Theta Flex, Sweden). The liquid droplets of 3 μL were placed in different areas on the surface. The value of CA was tested with an average of three independent measurements at least. Advancing and receding CAs were also obtained by CA meter (Biolin, Theta Flex, Sweden) while the probe fluid was added to and withdrawn from the drop, respectively. Each value reported is the average of at least 5 independent measurements. The values of contact angle hysteresis were obtained from three measurements. Ellipsometric thicknesses were measured using a Gaertner Scientific Stokes ellipsometer at a 70° incident angle (from the normal to the plane). The light source is a He-Ne laser ($\lambda = 632.8$ nm). Thickness was calculated using the manufacturer's GEMP software with the following refractive indices: air, $n_0 = 1$; silicon oxide and polymer layers, $n_1 = 1.46$; silicon substrate, $n_s = 3.85$ and $k_s = -0.02$ (absorption coefficient). Measurement error is within 1 Å as specified by the manufacturer. Thickness values were measured after surfaces were rinsed thoroughly with water, isopropanol, and toluene. Each value reported is the average of values determined at four different positions on each surface. The error bars for the thickness are standard deviations from four independent measurements. The surface roughness was measured by a confocal laser scanning microscope (Zeiss, LSM 900, Germany). Photographs and movies were taken by the camera (Canon, EOS R50, Japan).

3. Results and discussion

In our design, we chose PMTFPS over the commonly used PDMS due to its several advantageous properties. This fluorinated polysiloxane maintains high polymer-chain flexibility akin to PDMS while having a significantly lower surface energy of 13.6 mN/m compared to PDMS's 22.8 mN/m, attributed to the fluorination of its side chains. Additionally, K103, a commercial perfluoropolyether lubricant, was selected over silicone oil due to its remarkable reduction in surface tension, decreasing from 20.7 to 15.0 mN/m [38,39]. Moreover, the adhesion strength between PMTFPS and K103 was observed to be notably higher than that between PDMS and silicone oil as evidenced by an improved work of adhesion of 36.4 from 30.5 mN/m. These comparative analyses highlight PMTFPS/K103 as a superior slippery coating system compared to the traditional PDMS/silicone oil.

Titanium alloys have long been favored in marine metal structures; however, they have been plagued by severe biofouling issues due to their inherent compatibility with biological organisms [40]. In our study, titanium alloy TA24 served as the model substrate material. The fabrication process of the PMTFPS/K103 coating system on TA24 can be outlined briefly as follows, as depicted in Fig. 1c. Initially, the TA24 substrate underwent activation via low-temperature plasma treatment to generate numerous hydroxyl groups, serving as active sites for PMTFPS grafting. Subsequently, the PMTFPS polymer brush layer was grown onto the surface via acid-catalyzed polycondensation of MTFPS monomers, following the methodology outlined by Wang and McCarthy [41]. Finally, the slippery coating system was formed by spraying K103 onto the PMTFPS polymer brush grafted surface.

Our initial investigation focused on the adsorption of MTFPS on hydroxylated TA24 substrate, employing a combination of experimental and theoretical approaches. Ellipsometry data unveiled the formation of a stable MTFPS layer upon immersing the hydroxylated substrate in an MTFPS solution. The thickness reached 5.9 nm within 30 s, with no further increase observed even after extending the soaking time to 90 min. These findings suggest that surface hydroxyl groups facilitated the adsorption of MTFPS molecules and subsequently prompted rapid multi-layer assembly of MTFPS through OH-terminated MTFPS chains forming hydrogen bonds. To delve deeper into the interaction between an MTFPS monomer and the hydroxylated TA24 surface, molecular dynamics

simulations were employed, using a fully optimized model (Fig. S2). The simulations revealed that MTFPS initially adopted a metastable adsorption configuration at the surface around 150 ps, followed by relaxation to a stable configuration around 600 ps. The bonding distance was measured at 2.435 Å, with the interaction energy calculated to be 5.8 ± 0.05 kcal/mol, consistent with typical hydrogen bond interactions [42].

We conducted a detailed study on the growth process of the PMTFPS polymer brush during the polymerization of MTFPS monomers. It is established that a polymer brush layer must be confined within a thickness of 10 nm to achieve liquid-like properties with optimal dynamic flow characteristics [43]. Fig. 2a illustrates the evolution of PMTFPS polymer brush layer thickness and grafting density over time. The thickness incrementally increased from 6 to 9 nm within 45 min before experiencing rapid growth. Therefore, the polymerization time needs to be meticulously controlled within 45 min. Grafting density is another critical parameter affecting the dynamic flow characteristics of the polymer brush [44], quantified by a dimensionless index Σ . This index denotes the number of chains that occupy an area that a free non-overlapping polymer chain would normally fill under the same conditions [45]. Depending on the Σ value, polymer brushes exhibit different states: i) the 'mushroom' state at $\Sigma < 1$, where grafted chains are widely separated and the underlying solid substrate is exposed; ii) the 'mushroom-to-brush' transition state at $1 < \Sigma < 5$, where chains are moderately packed and capable of free rotation; and iii) the 'stretched brush' state at $\Sigma > 5$, where chains are densely packed and behave rigidly [46]. Only the 'mushroom-to-brush' transition state ($1 < \Sigma < 5$) is conducive to creating a liquid-like layer. Molecular dynamics simulations were utilized to explore the evolution of Σ with polymer brush thickness, predicting Σ values of 1.66, 2.01, 2.18, 2.96, and 6.69 for thicknesses of 5.9, 7.3, 7.6, 8.7, and 13.3 nm, respectively (Fig. S3). To achieve a liquid-like layer with moderate PMTFPS polymer brush density, we selected a thickness of 7.6 nm, corresponding to a polymerization time of 30 min, for subsequent characterization and incorporation of lubricant. CLSM revealed an increase in surface roughness by approximately 7 nm after PMTFPS polymer brush growth, consistent with ellipsometry-measured thickness (Fig. S4). Furthermore, high-resolution field-emission scanning electron microscopy (SEM) indicated that the PMTFPS polymer brush layer had a smooth surface (Fig. S5), conducive to its slippery performance.

As depicted in Fig. 2b, the PMTFPS polymer brush's glass-transition temperature (T_g) was -139°C as tested by differential scanning calorimetry. When the T_g of polymer is significantly lower than the room temperature, typically below -100°C , the grafted polymer layer exhibits liquid-like and flexibility characteristics [43].

The chemical bonding between the PMTFPS polymer brush layer and the TA24 substrate was investigated utilizing high-resolution XPS spectra (Fig. S6). Analysis of the Si 2p, Ti 2p, and O 1s spectra revealed the presence of Ti—O—Si bonds, confirming the covalent attachment of PMTFPS chains to the surface via dehydration condensation reactions between Si-OH and Ti-OH groups [47,48]. The binding strength of the PMTFPS polymer brush onto the substrate, serving as the physical basis for system stability, was assessed through ramp load scratch experiments. As depicted in Fig. 2c, the PMTFPS polymer brush exhibited markedly different surface mechanical properties compared to the unmodified substrate. The mechanical curve displayed an elastic stage at a normal load below 131.8 μN , indicative of the flexibility of the PMTFPS polymer brush. Complete failure of the PMTFPS polymer brush layer occurred when the normal load reached 272.2 μN . Considering the probe radius (0.85 μm) and the area density of polymer chains ($\Sigma = 2.18$), it was estimated that approximately 2.5×10^5 chains were present under the probe. Consequently, the average force experienced by a single PMTFPS chain at failure was calculated to be 530 pN. This critical force value aligns with the typical range required to break a covalent bond (usually ranging from 200 pN to a few nN) [49], providing further evidence of the covalent attachment of the PMTFPS polymer brush to

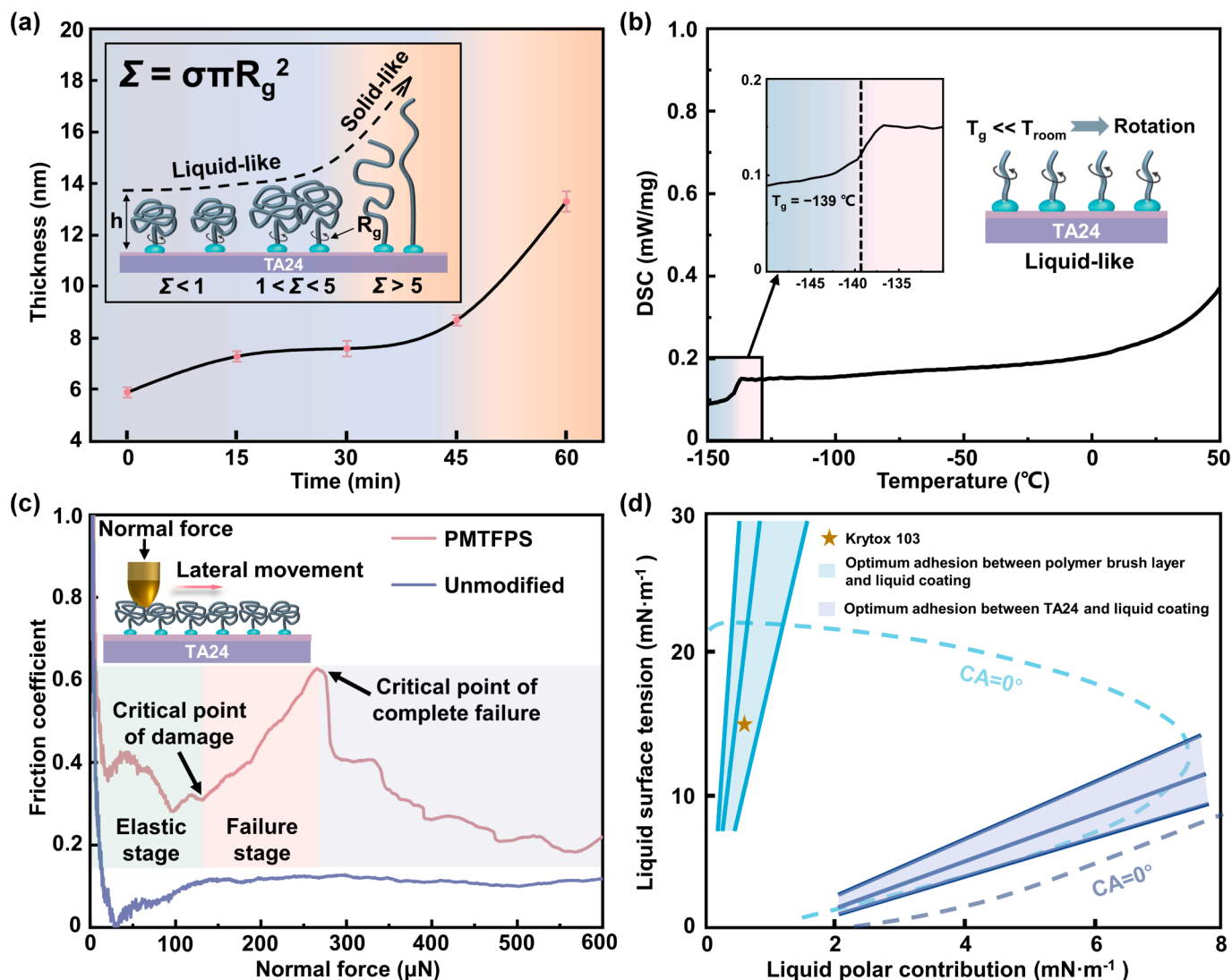


Fig. 2. Characterization of PMTFPS polymer brush layer and PMTFPS/K103 coating system. (a) Changes in the PMTFPS polymer brush layer thickness with grafting time (inset: transition of polymer brush properties with grafting density). (b) The glass-transition temperature (T_g) of the PMTFPS polymer brush. (c) Nanoscratch curves of unmodified TA24 and PMTFPS polymer brush grafted surfaces. The load was increased linearly with a rate of 60 $\mu\text{N}/\text{s}$ to a maximum of 600 μN . (d) An OWRK analysis of the adhesion of lubricant on PMTFPS polymer brush grafted surface (taking the adhesion on unmodified TA24 as a control).

the surface of TA24.

In a polymer brush/lubricant coating system, the interfacial adhesion between the polymer brush grafted surface and the lubricant plays a pivotal role in ensuring the coating system's stability. The OWRK method is typically employed to calculate the interfacial adhesion between solid and liquid interfaces. However, when applying the OWRK method in polymer brush/lubricant coating systems, it is still necessary to consider the interface issue. The polymer brush's grafting density is one of the key factors influencing the interfacial issue within the coating system. A.N. Harper and G. Liu first reported a theoretical model for the polymer brush/lubricant system's interface [50]. After injecting the lubricant into the polymer brush grafted surface with high grafting density, the lubricant will penetrate and swell the polymer brush layer, eliminating a distinct solid-liquid interface. Thus, the theoretical model with a three-layer structure including neat lubricant layer, brush layer, and lubricant-swollen crosslinked matrix was constructed. This theory provides precise design guidance for the microscopic interactions between interfaces in polymer brush/lubricant coating systems. While the polymer brush's grafting density on the metal surface may not be sufficiently high, it still meets our needs for the polymer brush layer's functional design. We hold the opinion that when the polymer brush

layer's grafting density is low, the polymer brush grafted metal surface can be regarded macroscopically as a chemically modified overall surface, and then the model has the possibility of being simplified to a double-layer structure including neat lubricant and polymer brush grafted surface.

To ensure interfacial stability, rational selection of lubricants was performed using work of adhesion analysis based on the OWRK method [51]. We investigated the Krytox series of perfluoropolyether lubricants, and K103 demonstrated favorable adhesion performance with the PMTFPS polymer brush. As illustrated in Fig. 2d, K103 fell within the corridor for optimal adhesion and exhibited spreading behavior within the 0° wetting envelope on the PMTFPS polymer brush layer. The work of adhesion between K103 and PMTFPS polymer brush grafted surface was calculated to be 36.4 mJ/m^2 .

In Fig. 3a, we observe the change in contact angle (CA) on the TA24 substrate following various modification processes. The CA of the TA24 substrate markedly decreased from $59.2^{\circ} \pm 0.6^{\circ}$ to $1.6^{\circ} \pm 0.1^{\circ}$ after plasma activation, attributable to the generation of abundant surface hydroxyl groups. Subsequent growth of the PMTFPS polymer brush rendered the surface hydrophobic, resulting in a high CA of $108.1^{\circ} \pm 2.2^{\circ}$ due to the inherently hydrophobic nature of PMTFPS. The

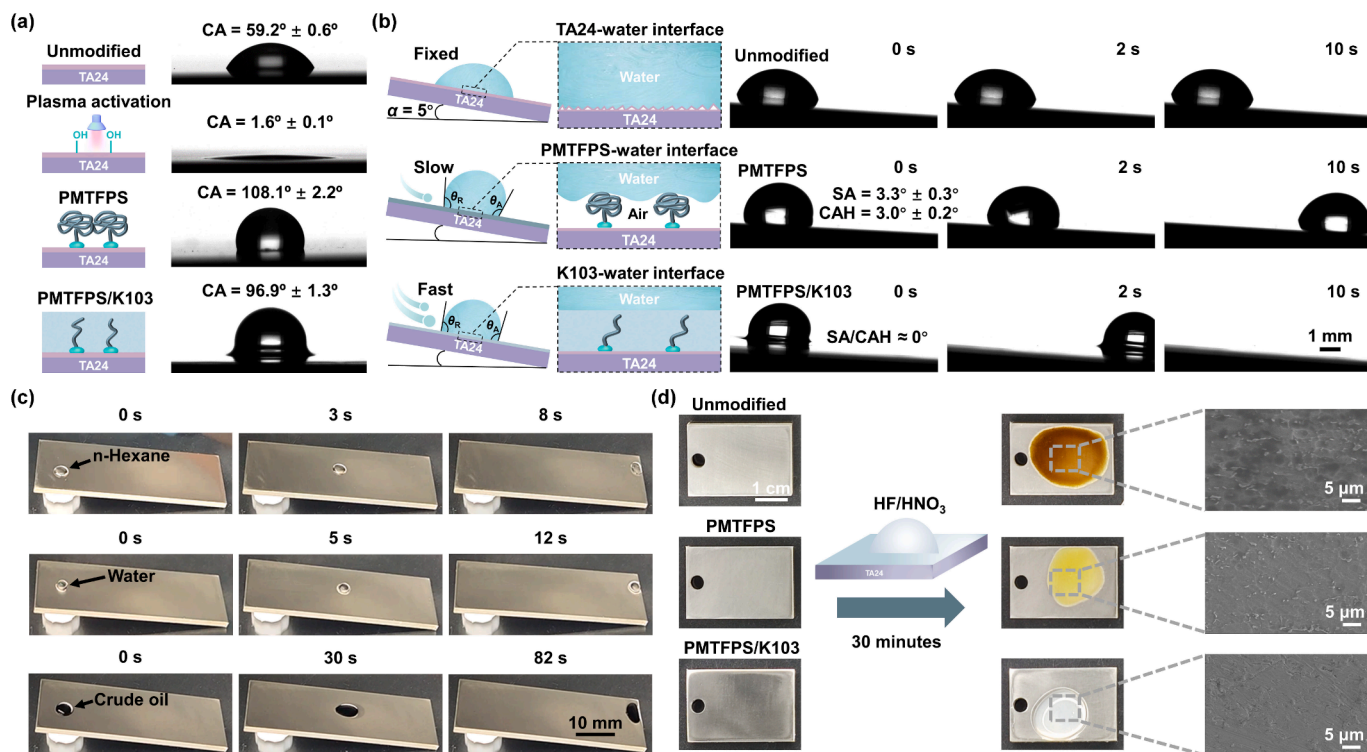


Fig. 3. Slipping, omniphobic, and anticorrosion properties of PMTFPS/K103 coating system. (a) Schematic illustration of different surfaces and their water contact angle values. (b) Schematic diagram illustrating the interface between various modified surfaces and water, and time-sequence images of a water droplet (15 μ L) sliding down at an angle of $\alpha = 5^\circ$. (c) Time-sequence images of the mobility of different probe liquids on PMTFPS/K103 surface. (d) Corrosion resistance tests of the unmodified, PMTFPS and PMTFPS/K103 surfaces.

incorporation of K103 further reduced the CA to $96.9^\circ \pm 1.3^\circ$, influenced by the lubricant layer. We proceeded to measure the contact angle hysteresis (CAH) and the sliding angle (SA), critical for evaluating the ease of initiating droplet motion and pivotal in liquid repellency. Fig. 3b illustrates the water droplets' slipping behavior on the three surfaces at a tilt angle (α) of 5° . On the pristine TA24 surface, the water droplet appeared immobilized, exhibiting a high CAH value of $42.0^\circ \pm 2.5^\circ$ due to its non-slippy nature. Conversely, the PMTFPS surface exhibited ultralow CAH and SA values of $3.0^\circ \pm 0.2^\circ$ and $3.3^\circ \pm 0.3^\circ$, respectively. The water droplet sliding down with a velocity of 2 mm/s. This exceptional slippy performance stemmed from the liquid-like nature of the PMTFPS polymer brush, facilitated by highly flexible polymer chains [52]. Upon modification with K103, the CAH and SA values were both further reduced to approximately 0° , and sliding became significantly faster with a velocity of 10 mm/s [53]. Thus, the PMTFPS/K103 surface showcased promising water-repelling and slippy properties.

Furthermore, the PMTFPS/K103 coating system demonstrated exceptional omniphobic properties. As depicted in Fig. 3c, it repelled not only water but also non-polar (n-hexane) and viscoelastic (crude oil) liquids. All tested liquids exhibited negligible CAH of approximately 0° on the surface. Upon application of n-hexane and crude oil droplets onto the surface, they instantaneously slid off without leaving any discernible wetting imprint, despite their low CA values [54,55].

Corrosion stands as a primary cause of metal component failure. Despite the corrosion resistance inherent in titanium alloys due to the presence of a nanoscale oxide protection layer, they remain susceptible to corrosion when exposed to severe environments such as strong acids [56,57]. To evaluate the anti-corrosion performance, we subjected the materials to an extremely corrosive agent comprising hydrofluoric acid, nitric acid, and deionized water in a volume ratio of 1:1:7, as depicted in Fig. 3d. After a 30-minute exposure to the corrosive agent, the unmodified TA24 substrate exhibited noticeable corrosion, characterized by the formation of brown products and a roughened surface, as observed

in SEM images. In contrast, for the PMTFPS surface, only light-colored products were observed, and the resulting surface remained significantly intact, indicating substantial corrosion mitigation attributed to the water-repellent nature of the liquid-like layer. However, it's important to note that polysiloxanes are susceptible to etching by hydrofluoric acid (HF), thus the PMTFPS polymer brush layer is unable to withstand the corrosive agent for an extended period. Conversely, the PMTFPS/K103 surface provided an excellent barrier against corrosion, as evidenced by the absence of any corrosion products or damage to the surface morphology. This remarkable corrosion resistance can be attributed to the water-repellency and chemical inertness of K103, which effectively prevented any contact between the corrosive solution and the TA24 surface.

To evaluate the stability of the PMTFPS polymer brush layer and PMTFPS/K103 coating system under dynamic shear flow [58], they were subjected to seawater scouring cycles (scouring at 4.9 m/s for one hour followed by immersion for another hour as one cycle), as depicted in Fig. 4a. The CA and CAH values were monitored after each scouring cycle (Fig. 4b and c). Continuous decreases in CA and increases in CAH were observed for both the PMTFPS polymer brush layer and the PMTFPS/K103 coating system, indicating shear-induced degradation. However, notably, the PMTFPS/K103 coating system exhibited a significantly slower reduction in surface repellency during the scouring cycles, indicating its enhanced stability compared to the PMTFPS brush layer. The static stability of the PMTFPS polymer brush and PMTFPS/K103 coating system in seawater was also tested as shown in Fig. 4d and e. They both showed promising liquid-repellency properties during a 30-day immersion cycle with only a slight decrease in CA values. The CAH values of the PMTFPS polymer brush and PMTFPS/K103 coating system surfaces could still be less than 10° and 5° , respectively [59]. As depicted in Fig. 4f, the surface friction coefficient was analyzed using cast iron HT200 with a normal force of 1 N applied. The PMTFPS/K103 coating system exhibited an exceptionally low friction coefficient value of 0.03,

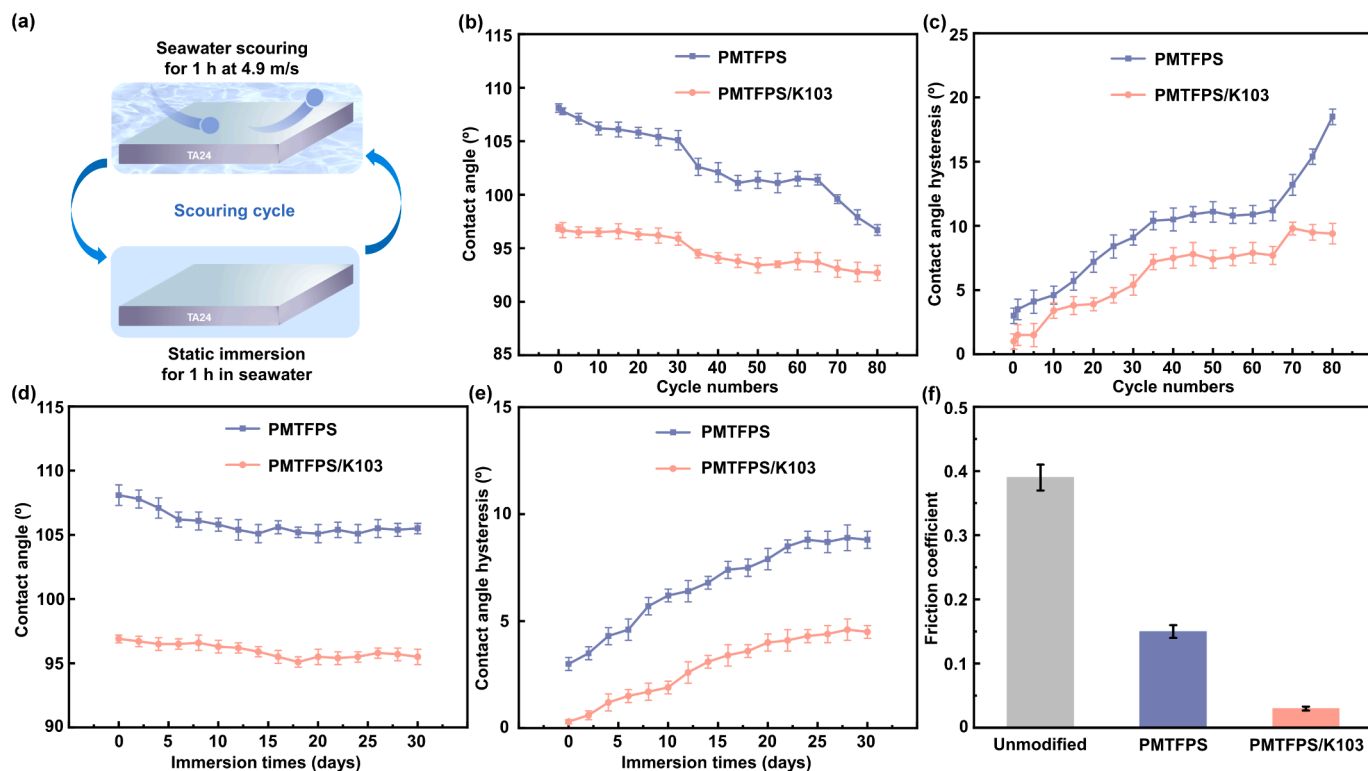


Fig. 4. Stability tests of PMTFPS polymer brush layer and PMTFPS/K103 coating system. (a) Schematic diagram of scouring cycle tests. (b-c) The changing trend of liquid repellency property of PMTFPS and PMTFPS/K103 surfaces for different scouring cycles. (d-e) The changing trend of liquid repellency property of PMTFPS and PMTFPS/K103 surfaces with different immersion times. (f) The friction coefficient tests of the unmodified, PMTFPS and PMTFPS/K103 surfaces.

representing an 80 % reduction compared to the PMTFPS polymer brush layer. This result indicates that K103 lubricant could assist the PMTFPS polymer brush in resisting shear action, as the lubricant acts as a buffer shielding the polymer from direct attack by water flow. Therefore, the physically adsorbed lubricant exhibits a synergistic effect with the

grafted polymer brush, enhancing the stability of the entire system.

The anti-biofouling efficacy of the system was evaluated through anti-diatom and anti-bacterial assays, employing *P. tricornutum* and *Bacillus sp.* as representative marine fouling microorganisms. The substrates were immersed in a solution of *P. tricornutum* (1×10^6 cfu/mL) or

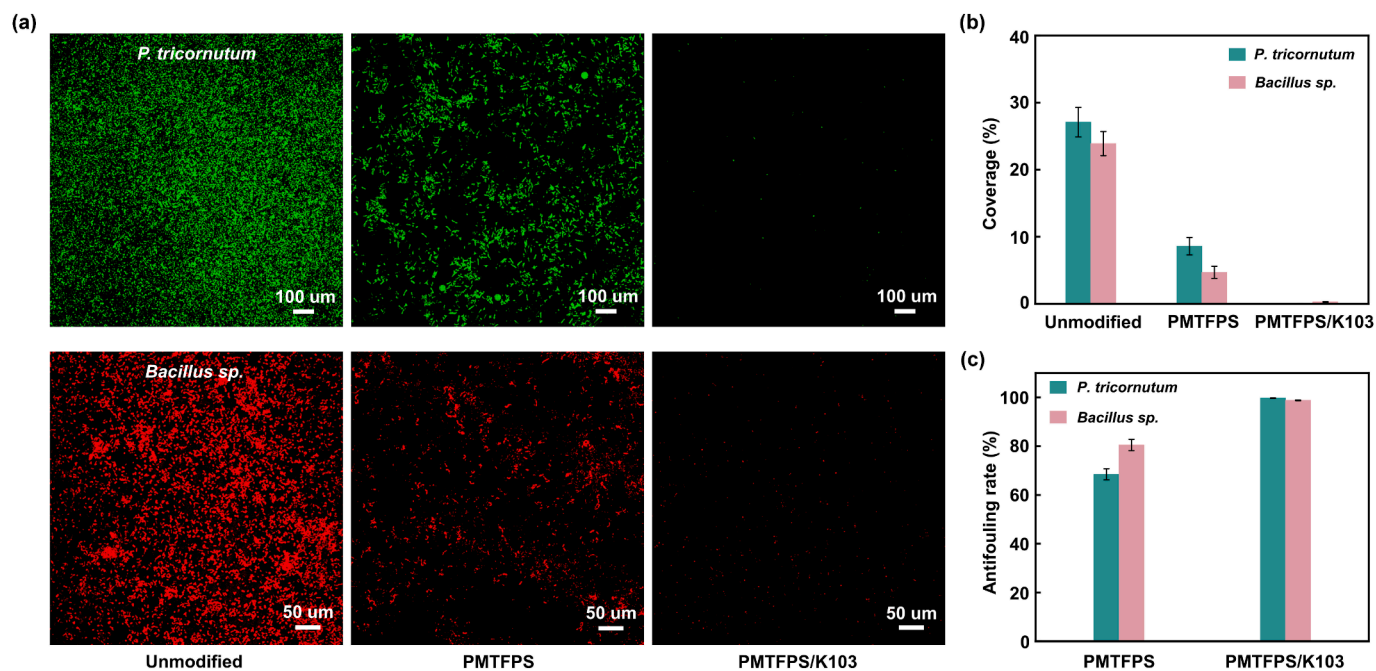


Fig. 5. Antibacterial and anti-diatom assays. (a) Fluorescence microscopy images of the unmodified, PMTFPS, PMTFPS/K103 surfaces after immersion with *P. tricornutum* and *Bacillus sp.*, respectively. (b) The corresponding coverage values of *P. tricornutum* and *Bacillus sp.* on the unmodified, PMTFPS and PMTFPS/K103 surfaces. (c) The antifouling rate of PMTFPS and PMTFPS/K103 surfaces for *P. tricornutum* and *Bacillus sp.*.

Bacillus sp. (5×10^6 cells/mL) for periods 7 days and 3 days, respectively. Subsequently, the substrates were thoroughly rinsed, and the attached microorganisms were fixed and stained on the surfaces. The attachment of the microorganisms on the substrates was visualized by CLSM fluorescence images (Fig. 5a), and the coverage was quantified based on averaged fluorescence intensity (Fig. 5b and c). Both microorganisms exhibited dense and uniform adhesion layers on the TA24 surface. The coverage became looser and more uneven on the PMTFPS surface, resulting in a $68.5 \pm 2.2\%$ and $80.5 \pm 2.3\%$ reduction in the coverage of *P. tricornutum* and *Bacillus* sp., respectively, indicating moderate anti-diatom/anti-bacterial performance of the PMTFPS polymer brush. In contrast, the surface of PMTFPS/K103 exhibited only sparse, isolated single cells or microcolonies, with antifouling rates reaching $99.75 \pm 0.05\%$ and $98.8 \pm 0.1\%$ for *P. tricornutum* and *Bacillus* sp., respectively. These results underscore the excellent resistance of the PMTFPS/K103 coating system against fouling by marine microorganisms.

We further assessed the antifouling performance under natural seawater conditions through a comprehensive marine field test conducted in the East China Sea near Xiamen (Fig. S7) over a period of 150 days. Photographs of the tested panels revealed varying degrees of biofouling on different surfaces over time (Fig. 6). Within the first 30 days, the unmodified TA24 surface exhibited evident fouling, while there was no sign of biofouling on the PMTFPS or PMTFPS/K103 surfaces. By day 90, the unmodified TA24 surface displayed severe biofouling, characterized by a dark coating composed of biofilms and macro-organisms, and the PMTFPS polymer brush grafted surface also exhibited noticeable fouling with the attachment of a few microalgae. In contrast, only minimal biofouling was observed on the PMTFPS/K103 surface at 90 days. Thus, the PMTFPS/K103 coating system effectively prevented significant marine biofouling for at least 90 days under static immersion. However, after 150 days of immersion, heavy fouling was evident on all three surfaces, characterized by the thick coverage of biofilms and the settlement of barnacles and shellfish on their surfaces.

It is crucial to highlight that submerged surfaces are often subjected to dynamic flow conditions rather than static immersion. To simulate dynamic antifouling performance, the surfaces exposed for 150 days were further subjected to rotational motion at a linear velocity of 4.9 m/s for 3 h in seawater. Following this shearing action by seawater, the unmodified TA24 surface exhibited minimal desorption of foulants, indicating the strong adhesion of marine organisms to the TA24 surface. In comparison, on the PMTFPS polymer brush grafted surface, a portion of the biofilm was peeled off, and some barnacles in the early stage of metamorphosis were removed, benefitting from the dynamic characteristics of the PMTFPS polymer brush. However, settled shellfish seemed unaffected by the shearing force on the PMTFPS polymer brush surface. In contrast, most of the biofilms were desorbed from the PMTFPS/K103 coating surface, exposing the substrate surface. Additionally, all barnacles and shellfish were removed from the surface, leaving behind only small calcification marks. These results not only demonstrate the exceptional antifouling ability but also the long-term durability of the PMTFPS/K103 coating system.

Our study on the PMTFPS/K103 coating system represents the first marine field test results for a polymer brush/lubricant system, as current research on PDMS/silicone oil coating systems for antifouling is predominantly limited to laboratory conditions. A literature review indicates that the antifouling performance of the PMTFPS/K103 coating system may surpass that of most slippery liquid coatings in long-term marine field tests (see a summary of the results in Table S3). For instance, silicone oil-infused PDMS networks (known as iPDMS) were visibly colonized by microbial biofilms within 20 days [60], while electrostatically flocced PDMS/silicone oil coatings exhibited dark-colored biofilms after 47 days of immersion [61]. Other silicone oil-infused surfaces, such as microchannel silica [62] and wrinkled Teflon [63], experienced a severe accumulation of foulants at 30 and 49 days, respectively. Recently, Rawlinson et al. developed a novel squalene-impregnated poly(styrene) nanocoating, reporting that 19 % of the coated surface was fouled at 42 days with the settlement of some

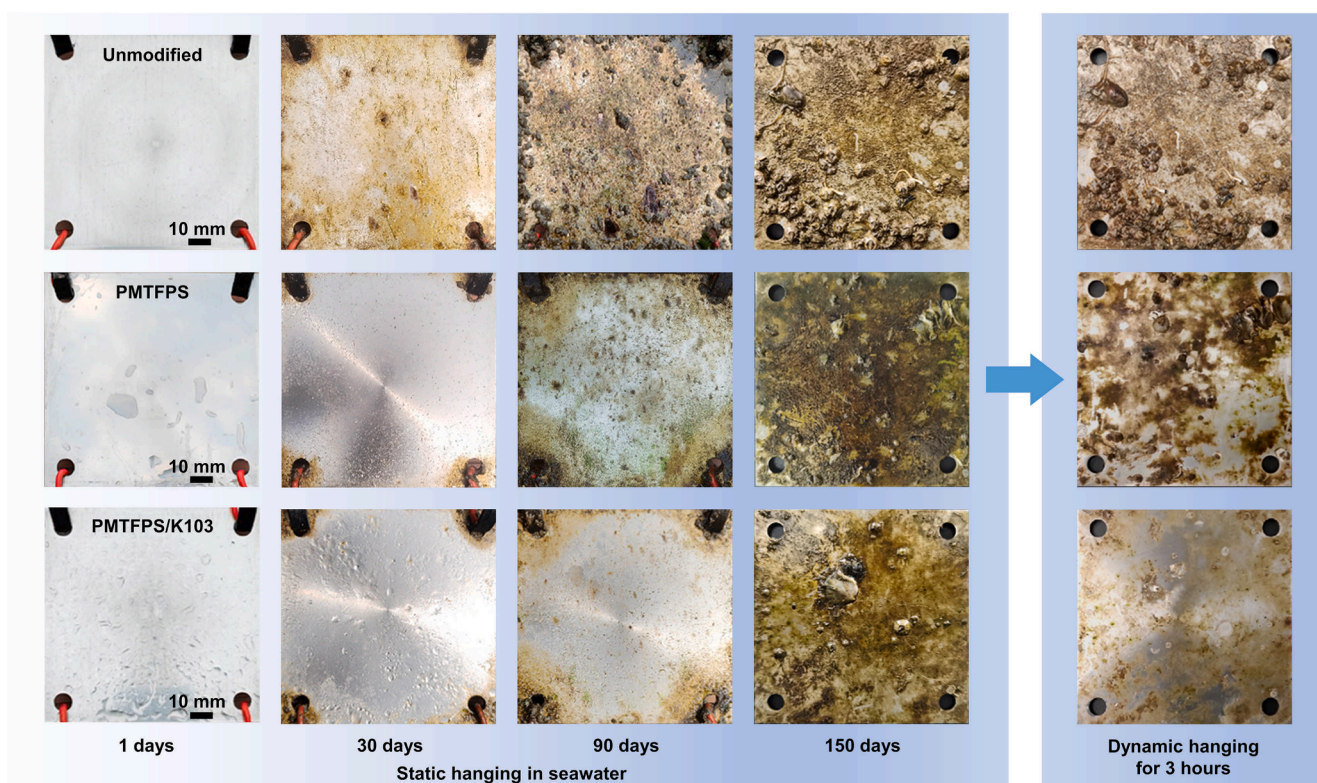


Fig. 6. Photographs of tested panels during the marine field test. The panels (100 mm × 100 mm) were fixed on the hanging bracket (top: Unmodified, middle: PMTFPS, bottom: PMTFPS/K103).

macrofoulers [64]. Furthermore, an assessment of some commercial slippery coatings revealed thick biofilm coverage at 20 days and severe macrofouling by barnacles and mussels at 74 days, as reported by the Miserez group [60].

4. Conclusion

In summary, we have demonstrated a novel and rationally designed slippery coating system of PMTFPS/K103 for marine antifouling applications. The design idea of the coating system in this work can serve as a supplement to the design theory of PDMS/silicone oil coating system. We successfully fabricated the PMTFPS/K103 coating system on TA24 surfaces, showcasing excellent surface omniphobicity and remarkable corrosion resistance to an extremely corrosive agent. Notably, the PMTFPS/K103 coating system exhibits enhanced durability compared to the PMTFPS brush layer under static immersion and dynamic scouring, underscoring the protective effect of the physically adsorbed lubricant on the covalently grafted polymer brush. The PMTFPS/K103 coating system achieved antifouling rates of approximately 98.8 % and 99.8 % for *P. tricornutum* and *Bacillus sp.*, respectively, indicating its exceptional performance in preventing the adhesion of marine microorganisms. Marine field tests further confirm negligible biofouling at 90 days under static immersion, surpassing the reported antifouling performance of most slippery coating systems. Moreover, fouling organisms adhering for 150 days can be readily desorbed by shear force in flowing seawater. These results show the promising potential of the PMTFPS/K103 coating system as an effective antifouling strategy. We anticipate that this work will inspire future endeavors in the rational design of polymer brush/lubricant systems – an underexplored research direction offering opportunities to create high-performance slippery surfaces for marine antifouling applications.

CRediT authorship contribution statement

Baoxin Wang: Writing – original draft, Methodology. **Hongxing Ye:** Validation, Investigation. **Baiyi Chen:** Writing – review & editing, Funding acquisition, Conceptualization. **Pengjiao Zu:** Investigation. **Guizhen Lu:** Investigation. **Li Ma:** Writing – review & editing, Supervision. **Haibing Zhang:** Investigation. **Mingxian Sun:** Investigation. **Yuanjie Li:** Investigation. **Hongyi Liu:** Investigation. **Jianhua Wu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.153568>.

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